

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: The atmosphere: energy processes (UV, IR, visible light) - Set 1

Mark scheme

AQA spec 3.2.1.1; The physical environment; textbook pp. 86-89

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• CO₂ rises from about 412 ppm in January to a peak of 416 ppm in May, falls to a minimum of 407 ppm in September, then rises again • During spring / summer, photosynthesis by plants (growing season) removes CO₂ faster than respiration releases it, so the concentration falls • In autumn / winter photosynthesis is reduced (leaf fall) but respiration and decomposition continue, so CO₂ rises • The atmosphere is in dynamic equilibrium: the concentration fluctuates around a mean; more land / vegetation in the northern hemisphere makes the cycle large	4
01.2	• Natural processes that add gases (respiration, combustion, volcanoes, denitrification) are balanced by processes that remove them (photosynthesis, dissolving, nitrogen fixation) • So the average composition stays the same over long timescales although the rates vary and concentrations fluctuate around a mean	2
01.3	<i>Any 2 from:</i> • Absorbs harmful UV / solar wind radiation in the stratosphere (ozone layer) • Greenhouse gases delay the escape of infrared, keeping the surface warm (33 °C warmer than without) • Atmospheric pressure allows liquid water to exist • Distributes heat from the tropics to higher latitudes by winds • Transports water vapour to areas that would otherwise be dry	2
01.4	• Troposphere • Stratosphere	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89; AQA spec 3.2.1.1.	
02.1	• 25 + 5 = 30%	1

Question	Marking guidance	Marks
02.2	• UV: short wavelength; mostly absorbed in the stratosphere by monatomic, diatomic and triatomic oxygen (ozone layer) which are formed and destroyed in a dynamic equilibrium; converted to chemical energy in these reactions; little reaches the surface • Visible light: transmitted through the atmosphere easily (some reflected by clouds / dust); absorbed by the Earth's surface and converted to heat; some absorbed by chlorophyll and converted to chemical energy in photosynthesis • IR: emitted by the warm surface (long-wave); absorbed by greenhouse gases (CO₂, water vapour, methane) and converted to heat; the warm atmosphere re-emits IR to the surface and reduces conduction losses; some escapes to space	6
02.3	• Greenhouse gases absorb most of the IR emitted by the surface and re-radiate it in all directions, including back down • The surface is repeatedly warmed and re-emits, so the total IR exchanged between surface and atmosphere exceeds the incoming solar energy; only the net loss (70%) balances the absorbed solar energy at equilibrium	2
02.4	• The atmosphere forms layers (troposphere, stratosphere, mesosphere) with different temperature gradients because energy is absorbed at different altitudes (eg UV heating the stratosphere, surface heating the troposphere)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87-89; AQA spec 3.2.1.1.	
03.1	• Most solar energy is absorbed at the surface in the tropics where insolation is most intense / the Sun is overhead • The warm surface heats the air above; warm winds (eg south-westerlies) carry heat energy to higher latitudes • Winds blowing over the oceans create currents (North Atlantic Conveyor / Gulf Stream) carrying warm water from the tropics to NW Europe • Evaporation in the tropics stores latent heat in water vapour which is released when it condenses at higher latitudes; also transports water to areas that would be dry	4
03.2	• The North Atlantic Conveyor brings warm water from the tropics to the coast of NW Europe • Prevailing south-westerly winds blowing over the warm sea carry the heat onto the land	2
03.3	• Atmospheric pressure controls how easily water molecules evaporate / escape from the surface • At low pressure water would boil / evaporate at low temperatures so all surface water would become vapour	2
03.4	<i>Any 2 from:</i> • Nitrogen • Oxygen • Carbon dioxide • Argon / neon / krypton / xenon	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87, 96-97; AQA spec 3.2.1.1, 3.2.2.3.	
04.1	• Oxygen	1

Question	Marking guidance	Marks
04.2	- A change to one process causes other processes to alter as a direct result (chains of linked processes / feedback) - There is still a lot that is not understood about how atmospheric processes work	2
04.3	- The surface warmed in the tropics heats the air above; warm winds (south-westerlies from the Caribbean) carry heat energy to higher latitudes - Winds carry water vapour (evaporated from warm oceans) to areas that would otherwise receive little precipitation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-88; AQA spec 3.2.1.1.	
05.1	$15 - (-18) = 33\text{ }^{\circ}\text{C}$	1
05.2	- Visible light passes easily through the atmosphere and is absorbed by the surface, which warms - The warm surface emits (long-wave / far) infrared radiation - Greenhouse gases (water vapour, CO_2, methane) absorb infrared better than the average for the atmosphere and convert it to heat - The warm atmosphere emits infrared which is absorbed by the surface, and it reduces heat loss from land and oceans by conduction	4
05.3	Any gas that is better at absorbing infrared radiation than the average for the atmosphere	1
05.4	- Arriving: mainly ultraviolet, visible light and near infrared (short wavelength); leaving: mainly long-wavelength (far) infrared - Gases that are transparent to the incoming short wavelengths can absorb the outgoing long wavelengths, so changing their concentration changes how much energy is retained	2
05.5	- The energy and the processes it drives control climate, ocean currents and the hydrological cycle - These determine the abiotic conditions (temperature, rainfall) that each species is adapted to, so a change alters where species can survive	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87-89; AQA spec 3.2.1.1.	
06.1	- Together they form a dispersed layer in the stratosphere (the ozone layer / ozonosphere) - They absorb ultraviolet light, which drives a dynamic equilibrium of reactions that form and destroy ozone - The UV energy is converted to chemical / heat energy so most UV that passed the upper atmosphere never reaches the surface	3
06.2	Any 2 from: - Provides carbon, oxygen, hydrogen and nitrogen (as N_2, O_2, CO_2, H_2O) for biological molecules - Delays the escape of infrared energy, keeping the surface warm - Distributes heat and water vapour by winds / ocean currents - Provides atmospheric pressure so liquid water can exist - Deflects damaging radiation of the solar wind in the upper atmosphere	2
06.3	Energy is absorbed at different altitudes (eg UV absorbed in the stratosphere, surface heating warms the troposphere) so temperature gradients change with altitude, producing layers	1

Question	Marking guidance	Marks
06.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Troposphere: enhanced greenhouse effect - anthropogenic CO₂ (fossil fuels, deforestation), methane (rice, livestock, landfill), N₂O, CFCs, tropospheric ozone; more infrared absorbed and re-emitted; aerosols and smoke increase albedo / scatter light; land-use change alters surface albedo and evapotranspiration • Stratosphere: CFCs / halons / HCFCs release chlorine and bromine which catalyse ozone destruction, shifting the O / O₂ / O₃ equilibrium; less UV absorbed in the stratosphere, more reaches the surface; stratospheric cooling • Consequences: changes to climate, ocean currents (thermohaline), hydrological cycle, sea level, species distributions; increased UV damage to organisms • Difficulties of prediction: linked processes where one change alters others; feedback (positive and negative) and tipping points; time lags (ocean heat capacity, long residence times of CFCs and CO₂); natural variability (solar output, volcanoes, Milankovitch cycles); incomplete understanding of atmospheric processes; limited historical data so proxies needed; models with different assumptions • Evaluation: some effects well understood (greenhouse physics, ozone chemistry proven, Montreal Protocol success) but scale and timing of regional impacts uncertain; precautionary principle • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89, 108-113; AQA spec 3.2.1.1, 3.2.1.3.	